

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Optimization in CS, Systems for Data Science, Advanced Machine Learning, Advanced NLP, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Publication

- “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages & Analysis	Python, SQL, TypeScript, JavaScript, pandas, NumPy, Excel / Google Sheets, Bash
AI Workflows & Agents	LLM agents, tool use, function calling, automation workflows, prompt design, evaluation methodology, OpenAI / Groq APIs, Ollama, MCP
Research & Analytics	RAG, RRF, cross-encoder reranking, MMR, knapsack token budgeting, batch experiments, holdout protection, llama-index, spaCy, sentence-transformers
Backend & Data Eng	FastAPI, Node.js, Express, REST APIs, Apache Spark / PySpark, Databricks, MongoDB, PostgreSQL, SQLite
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Research & Projects

AutoEval - Automation Agent for Internal Workflow Optimization

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an **automation agent** that proposes new queries and labels, scores them against multiple reference systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective, codifying what works as **reusable templates** and report cards committed back to Git.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and Git-backed acceptance gates so every accepted change ships with an auditable report card, demonstrating the AI-workflow pattern of taking a manual process and turning it into a reusable, measurable system.

AgenticSearch - LLM Pipeline for Diligence on Unstructured Sources

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **LLM-powered service** that mines structured entity tables from messy web sources via typed retrieval planning, page rerank, deterministic parsers with LLM fallback, output verification, and **cell-level provenance** for every result, the same shape of problem as turning fragmented protocol/market data into structured analyst-ready views.
- Hardened the service with **150+ automated tests**, provider routing/fallback across LLM backends, and pipeline instrumentation; reviewer-facing trust UI surfaces source URLs and confidence so every extracted fact is auditable.

KG2RAG-Enhanced - Research Methodology with Batch Experiments

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, llama-index, spaCy, PyTorch | UMass CS 685 Advanced NLP

- Extended KG2RAG with multi-view seed retrieval and a **0–1 knapsack token-budget** formulation (value = relevance × coverage) for evidence selection on HotpotQA; ran **100–500-question batch experiments** via an Ollama + LLaMA-3 pipeline with token and latency logging, demonstrating rigorous research methodology and analytical thinking.

Experience

Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected a full-stack **internal workflow tool** (React, Node.js, MongoDB) used daily by the AI team to ingest, label, and centralize training data; replaced fragmented spreadsheet workflows with role-based APIs and reviewer UIs, cutting manual operations by **100+ hours/month** and lifting throughput **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases, remediated **20+ VAPT findings**, and contributed to **\$340K+** in revenue.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.